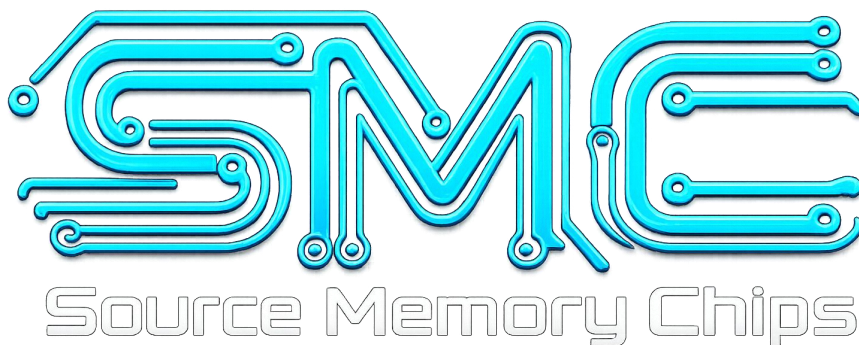


CXDB6CCDM

CXMT RAM

8GB LPDDR4X SDRAM

Technical Datasheet



Source Memory Chips (SMC) provides technical reference documentation for memory products used in mobile, portable and embedded electronic applications.

<https://www.sourcememorychips.com>

CXMT RAM — LPDDR4X SDRAM

Product Specification

Product Highlights

 8GB Capacity 64Gb mobile DRAM density	 LPDDR4X Low-power mobile SDRAM
 4266 Mbps Maximum rated data rate	 200-ball FBGA Compact discrete package

1. Product Overview

The CXMT CXDB6CCDM-MA-M is an 8GB LPDDR4X mobile SDRAM device supplied in a 200-ball discrete FBGA package. CXDB6CCDM is the base product identifier; the complete ordering code should be verified for design, qualification and procurement.

The device provides a dual-channel x32 data path ($2 \times x16$), two chip-selects and a rated data transfer rate of 4266 Mbps. Its low-voltage interface and compact package make it suitable for evaluation in space-constrained mobile and embedded designs. Final platform suitability must be confirmed against the complete ordering code and current manufacturer documentation.

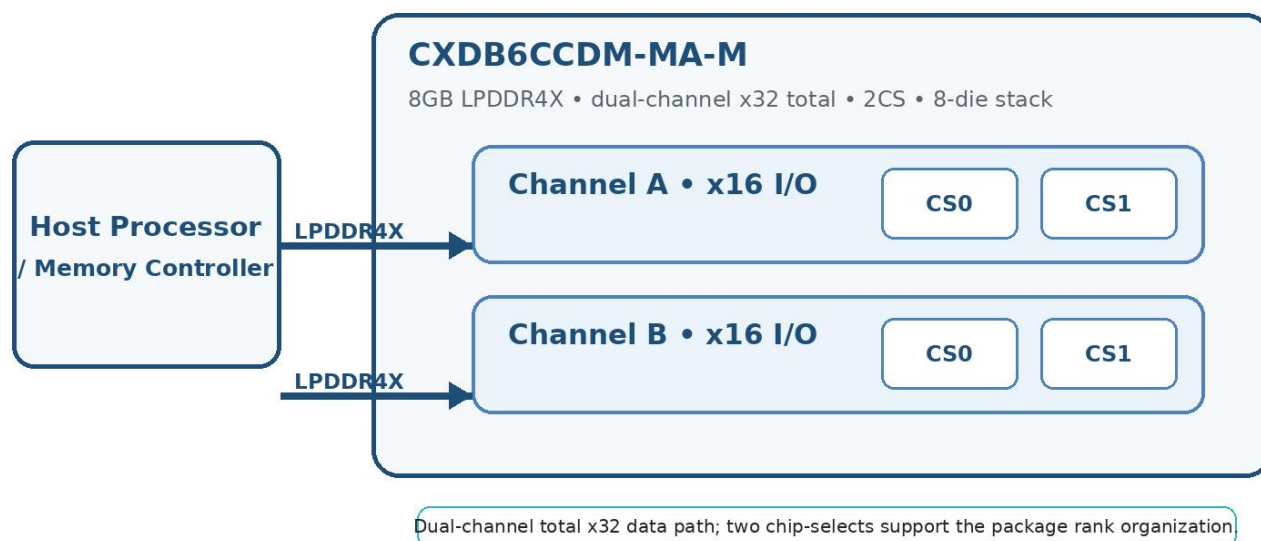
Item	Specification
Manufacturer	ChangXin Memory Technologies (CXMT)
Content Model	CXDB6CCDM
Full Ordering Part	CXDB6CCDM-MA-M
Product Type	8GB LPDDR4X Mobile SDRAM
Organization	Dual-channel x32 total ($2 \times x16$), 2CS
Data Rate	4266 Mbps
Package	200-ball discrete FBGA

2. Memory Architecture

CXDB6CCDM-MA-M uses an octo-die LPDDR4X architecture with two independent x16 channels, forming a total x32 data interface. Each channel provides separate clock, command/address, data-strobe and data connections to the host memory controller.

Memory Block	Function
LPDDR4X Channel A	x16 bidirectional data interface with independent clock, CA, DQS and CS signals.
LPDDR4X Channel B	x16 bidirectional data interface with independent clock, CA, DQS and CS signals.

CXDB6CCDM LPDDR4X Memory Architecture



3. Product Advantages

- 8GB high-density mobile memory in one compact package
- Dual-channel x32 organization for high memory bandwidth
- LPDDR4X low-voltage I/O for power-sensitive platforms
- Rated data transfer speed up to 4266 Mbps
- Configuration supports evaluation for compact, power-sensitive mobile and embedded system designs

4. General Specifications

Parameter	Specification
Manufacturer	ChangXin Memory Technologies (CXMT)
Full Part Number	CXDB6CCDM-MA-M
Product Type	Mobile LPDDR4X SDRAM
Density	8GB (64Gb)
Organization	2CH x32 total (2 × x16), 2CS, 8-die stack
Rated Data Rate	4266 Mbps
Package	200-ball discrete FBGA
Operating Temperature	Commercial grade: -25°C to +85°C
Environmental Status	Lead-free, halogen-free and RoHS compliant

Core Configuration

 8GB LPDDR4X density	 2CH x32 Dual-channel I/O	 2CS Dual-rank organization	 4266 Mbps Rated data rate	 200-ball Discrete FBGA
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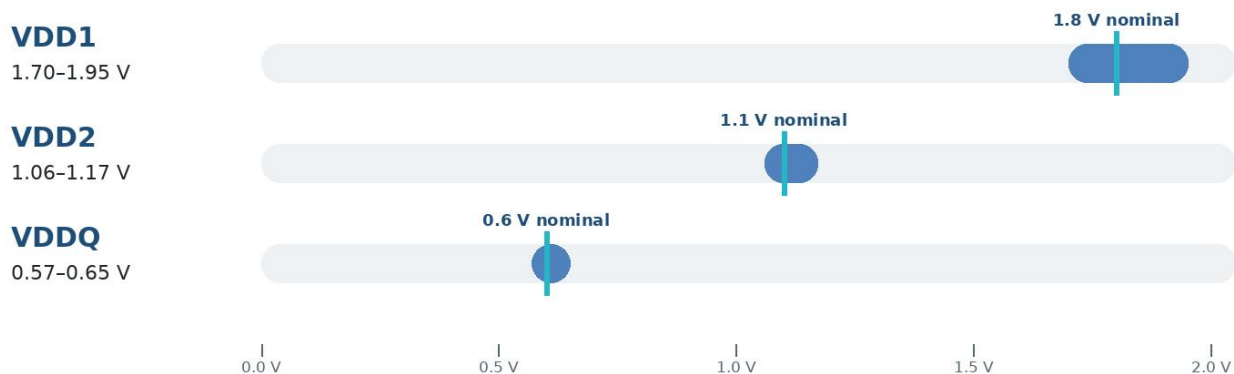
LPDDR4X Feature Support

Category	Supported Functions
Architecture	Dual-channel x32 total I/O, two chip-selects and octo-die package architecture
Low-Power Operation	LPDDR4X low-voltage I/O, self-refresh and power-down operating modes
Signal Interface	Differential clocks and strobes, data-mask inversion, ODT and ZQ calibration
Training & Control	Command-bus training, write leveling, read-preamble training and mode-register control
Reliability Support	Reset, temperature monitoring, refresh-rate management and timing derating support

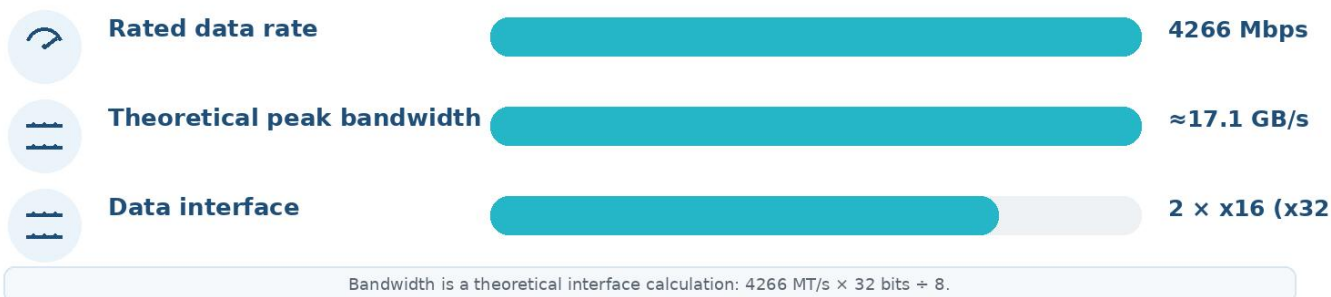
5. Electrical Characteristics & Performance

CXDB6CCDM-MA-M uses separate LPDDR4X power rails for the DRAM core, input buffers and I/O interface. The charts below summarize the recommended voltage windows and key interface-performance characteristics.

Recommended LPDDR4X Operating Voltages



Interface Performance Highlights



The maximum data rate is specified as 4266 Mbps. The approximate 17.1 GB/s figure is a theoretical peak interface calculation based on a total x32 data path and does not represent guaranteed application bandwidth.

6. Package Information

CXDB6CCDM-MA-M is supplied in a 200-ball discrete FBGA package. The simplified illustration below is provided for product-identification and mechanical-reference purposes only and must not be used as a PCB ball map or land-pattern drawing.

200-ball Discrete FBGA — Simplified Mechanical Reference

200 balls

Discrete FBGA

Pitch

0.8 mm X / 0.65 mm Y

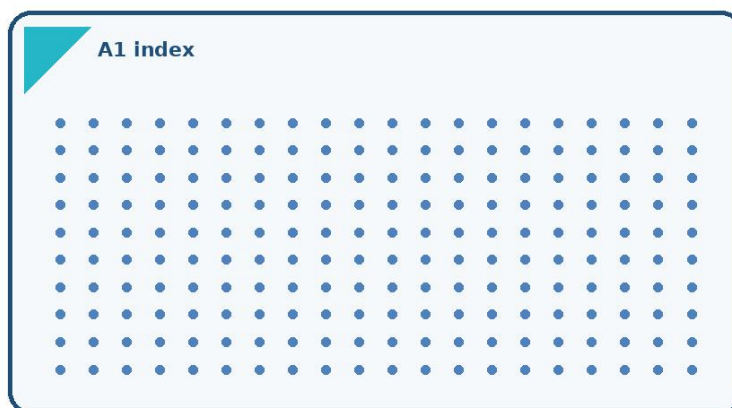
Thickness

Approx. 0.78 mm

Status

Lead-free / Halogen-free

NOT A BALL MAP



Reference body size: approximately 15.0 × 10.0 mm

Mechanical Item	Specification
Package Type	200-ball discrete FBGA
Reference Body Size	Approximately 15.0 mm × 10.0 mm
Reference Thickness	Approximately 0.78 mm
Ball Pitch	0.8 mm (X axis) / 0.65 mm (Y axis)
Orientation	Top view with A1 index at the upper-left corner

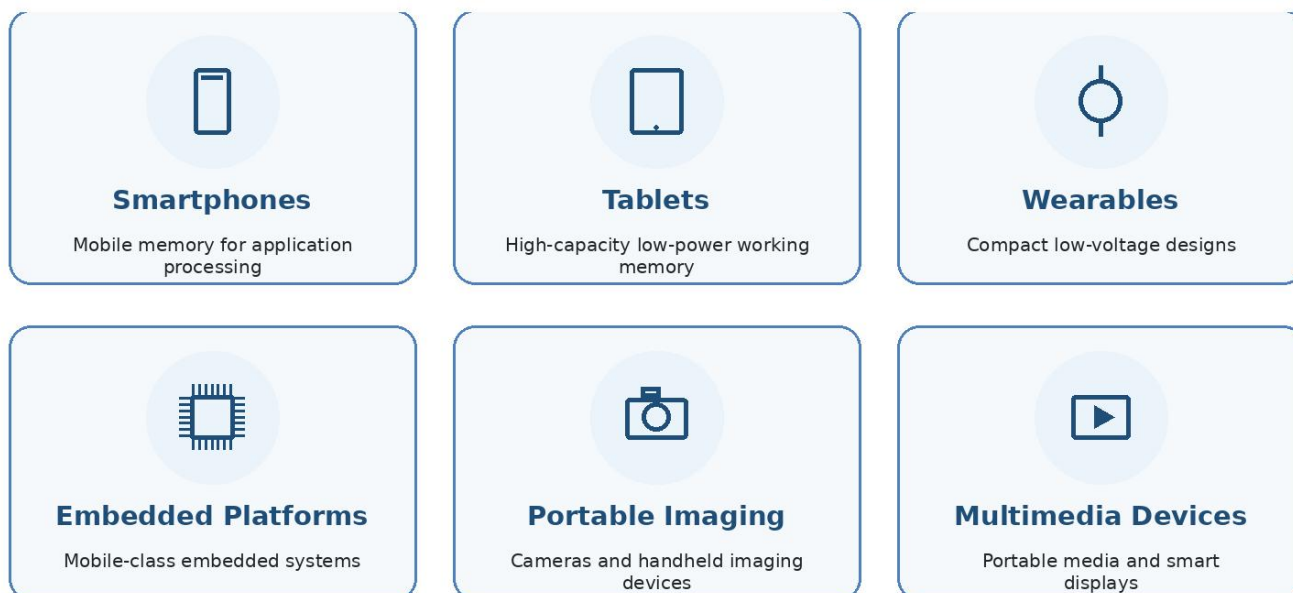
Package and Assembly Notes

- Use the official 200-ball ballout and A1 index to confirm package orientation.
- Verify the complete CXMT package drawing before creating the PCB land pattern.
- Confirm X/Y ball pitch, solder-mask definition, reflow profile and inspection criteria with the assembler.
- Follow the official power, signal-integrity and routing guidance for all LPDDR4X channels and supply rails.

7. Typical Application Examples

Based on its documented LPDDR4X configuration, CXDB6CCDM-MA-M may be considered for compact electronic systems requiring high-capacity, high-bandwidth and low-power working memory. The examples below are SMC reference use cases inferred from the device configuration and are not manufacturer-qualified end-use designations.

Typical Application Examples - SMC Reference



Reference Application Summary

Reference Application Area	Memory Requirement	CXDB6CCDM Advantage
Smartphones	High-capacity, low-power working memory	8GB LPDDR4X with dual-channel x32 interface
Tablets / Handhelds	High bandwidth in a compact footprint	4266 Mbps rated data rate and 200-ball FBGA
Wearables / Portable Devices	Low-voltage memory operation	LPDDR4X 0.6V nominal I/O supply
Embedded Mobile Platforms	Integrated high-density DRAM	Compact 8GB discrete memory solution

8. Ordering Information & Disclaimer

Ordering Information

Part Number	Manufacturer	Product Type	Package
CXDB6CCDM-MA-M	CXMT	8GB LPDDR4X SDRAM	200-ball FBGA

Product Identification

Item	Description
Content Model	CXDB6CCDM
Full Ordering Part	CXDB6CCDM-MA-M
Product Category	CXMT RAM - Mobile LPDDR4X SDRAM
Reference Application Scope	Mobile, portable and embedded systems (SMC reference scope)

Revision History

Revision	Date	Description
Rev 1.0	July 2026	Initial SMC technical-reference datasheet release

Disclaimer

Source Memory Chips (SMC) provides this document for technical reference and product-identification purposes only. It is an independent SMC reference document and is not an original CXMT manufacturer datasheet or a substitute for official design documentation.

Specifications, package information, operating conditions and availability may change without notice. Customers must verify the complete device marking and use current official manufacturer documentation for final design, qualification and procurement decisions.

Technical basis: a publicly available technical datasheet copy attributed to CXMT for CXDB6CCDM-MA-M, Version 1.0, Aug. 31, 2022. Package dimensions are additionally cross-checked against independent public package analysis and are presented as reference values.

Contact Information

Source Memory Chips (SMC) | Website: www.sourcememorychips.com